

MUHAMMAD ASMAN

Backend Engineer | Go · NestJS · .NET · Microservices

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SUMMARY

Backend engineer with 3 years of experience building microservices in Go, NestJS, and .NET. Currently Head of Backend Development Section at Ganesha Operation, setting technical direction across dozens of services while staying hands-on with critical production issues and major features. Strong in gRPC, RabbitMQ, WebSocket, and asynchronous processing, with experience translating business requirements into maintainable backend solutions.

EXPERIENCE

Head of Backend Development Section — Ganesha Operation May 2026 – Present
Bandung | Promoted from Backend Developer

- **Set technical direction** for backend development across multiple services and business modules, while staying hands-on with critical production issues and major features.
- **Work across dozens of services** in a microservices environment (Go, NestJS) backed by PostgreSQL, Redis, and RabbitMQ.
- **Consolidated two real-time game services** (Socket.IO and native socket) into a single architecture built on native WebSocket.
- **Built Go-based background processing** for large, non-real-time datasets spanning multiple services, run as scheduled jobs.
- Design and maintain REST and gRPC APIs, asynchronous workflows, and real-time services.
- Work with engineering, business, and management stakeholders to clarify requirements and turn business processes into backend solutions.

Backend Developer — Ganesha Operation Nov 2024 – May 2026

- Developed backend services in Go and NestJS within a microservices architecture, integrating services over REST and gRPC.
- Delivered backend modules for leave management, student management and progress reporting, marketing and sales data, payments, and other operational processes.
- Contributed to API design, database integration, and architecture decisions; investigated and resolved issues across development and production environments.

Frontend Developer (Freelance) — Castellum Digital Indonesia Sep 2024 – Oct 2024

- Built a responsive React.js dashboard for the SATUSEHAT program; cut page load time by 40% and raised user engagement by 25%.

Software Engineer — Refactory Sep 2023 – Sep 2024

- Built middleware services and APIs in .NET to integrate enterprise procurement systems (BSI KTB), and backend APIs in Go for high-performance web applications such as the Tempo news portal.
- Contributed to the design and integration of microservices-based enterprise systems (procurement, finance, vendor management) using Clean Architecture.
- Developed Vue.js and Razor interfaces, a Flutter mobile app (BLoC), and automated tests with Robot Framework.

Freelance Software Engineer — Jerivo Jan 2023 – Present

- Designed and delivered software solutions to client requirements across backend, web, and application development, including implementation, debugging, and maintenance of production applications.

TECHNICAL SKILLS

Languages: Go, TypeScript, C#
Frameworks: NestJS, .NET
Architecture: Microservices, Distributed Systems, Clean Architecture
APIs & Communication: REST, gRPC, WebSocket
Data & Messaging: PostgreSQL, Redis, RabbitMQ, MySQL
Tools: Docker, Git, Robot Framework
Also familiar with: React.js, Vue.js, Flutter (BLoC)

EDUCATION

Universitas Alma Ata, Yogyakarta — Bachelor of Computer Science (S.Kom), Informatics
GPA 3.83 / 4.00

Sep 2019 – Aug 2023